

Synoptic CB: Macrophyte Fluxes

2024 Samples

2026-02-23

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#Read in and collate all the files in the Processed Data folder

```
ch4 <- read.csv("Processed Data/COMPASS_Synoptic_CB_Macrophyte_Fluxes_Final_2024_CH4.csv")
ch4$X <- NULL
ch4 <- ch4 %>%
  rename_with(
    .cols = 7:(ncol(.) - 3),
    .fn = ~paste0("ch4_", .)
  )
```

```
co2 <- read.csv("Processed Data/COMPASS_Synoptic_CB_Macrophyte_Fluxes_Final_2024_CO2.csv")
co2$X <- NULL
co2 <- co2 %>%
  rename_with(
    .cols = 7:(ncol(.) - 3),
    .fn = ~paste0("co2_", .)
  )
```

Combine all files

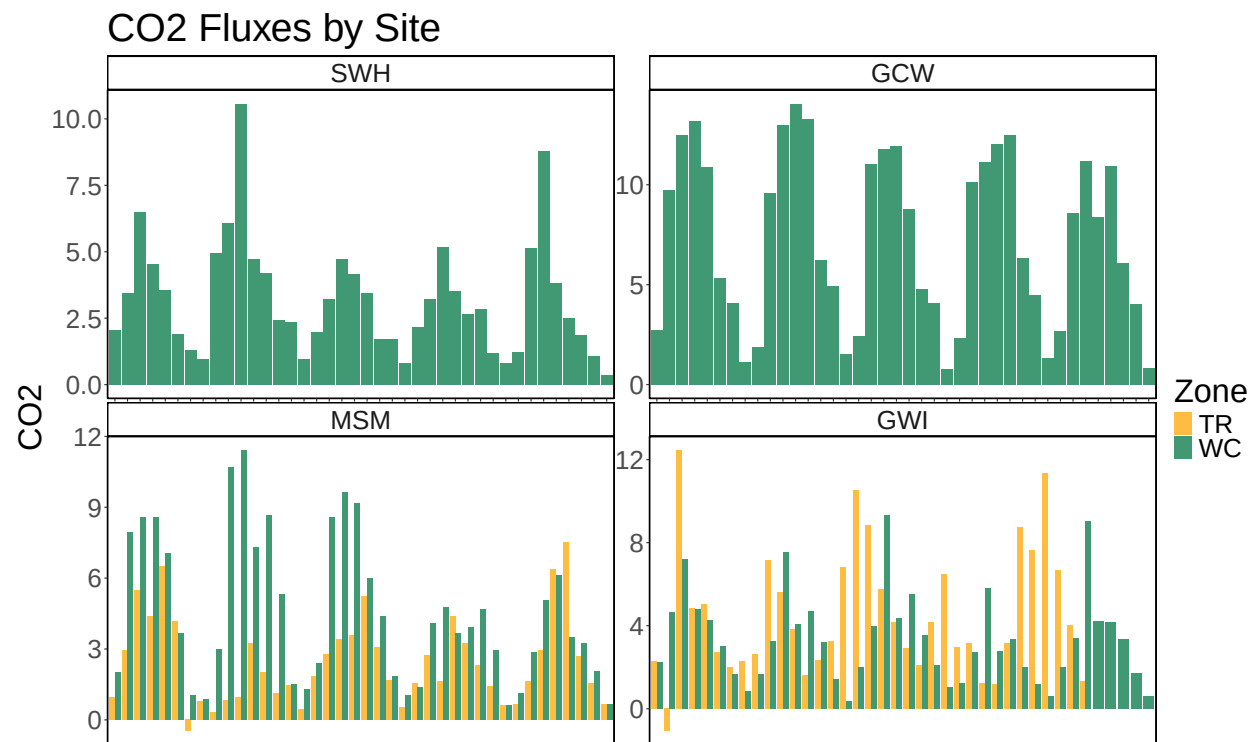
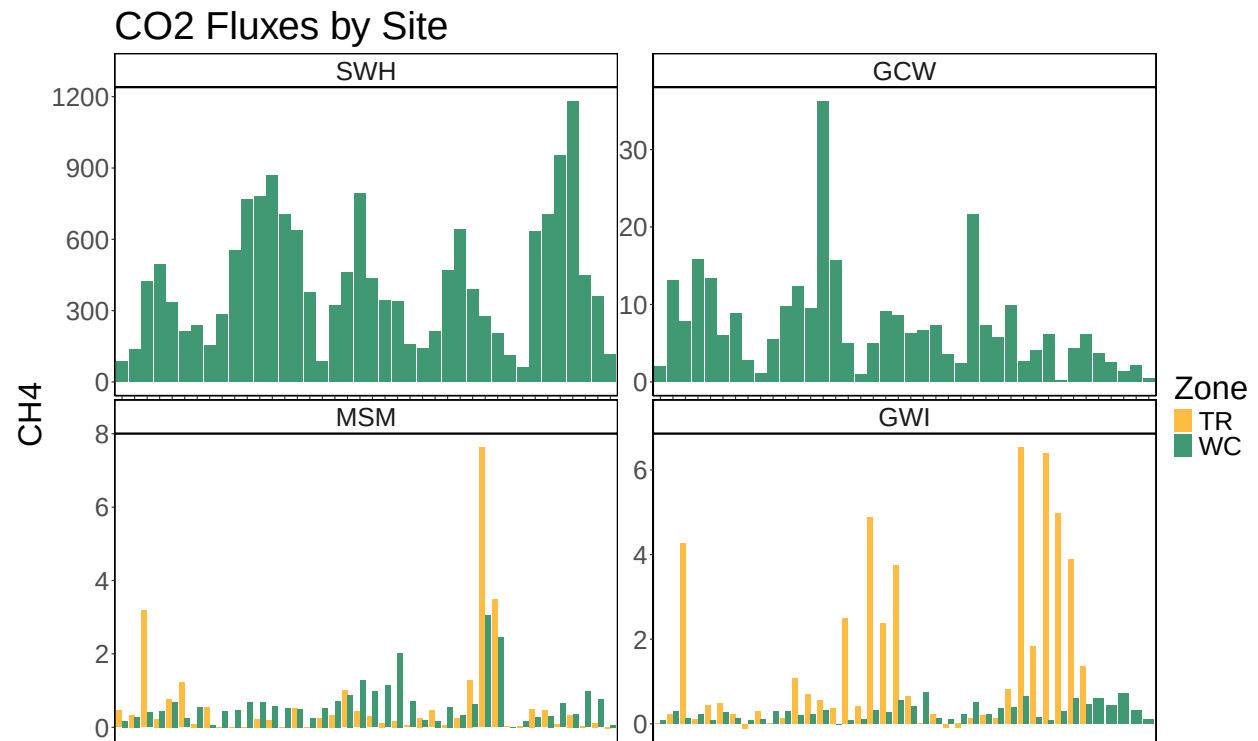
```
all_dat <- merge(ch4, co2, by = c("Plot", "Site", "Zone", "Replicate", "Date", "TIMESTAMP", "TIMESTAMP_1"))
```

##Rename Zone

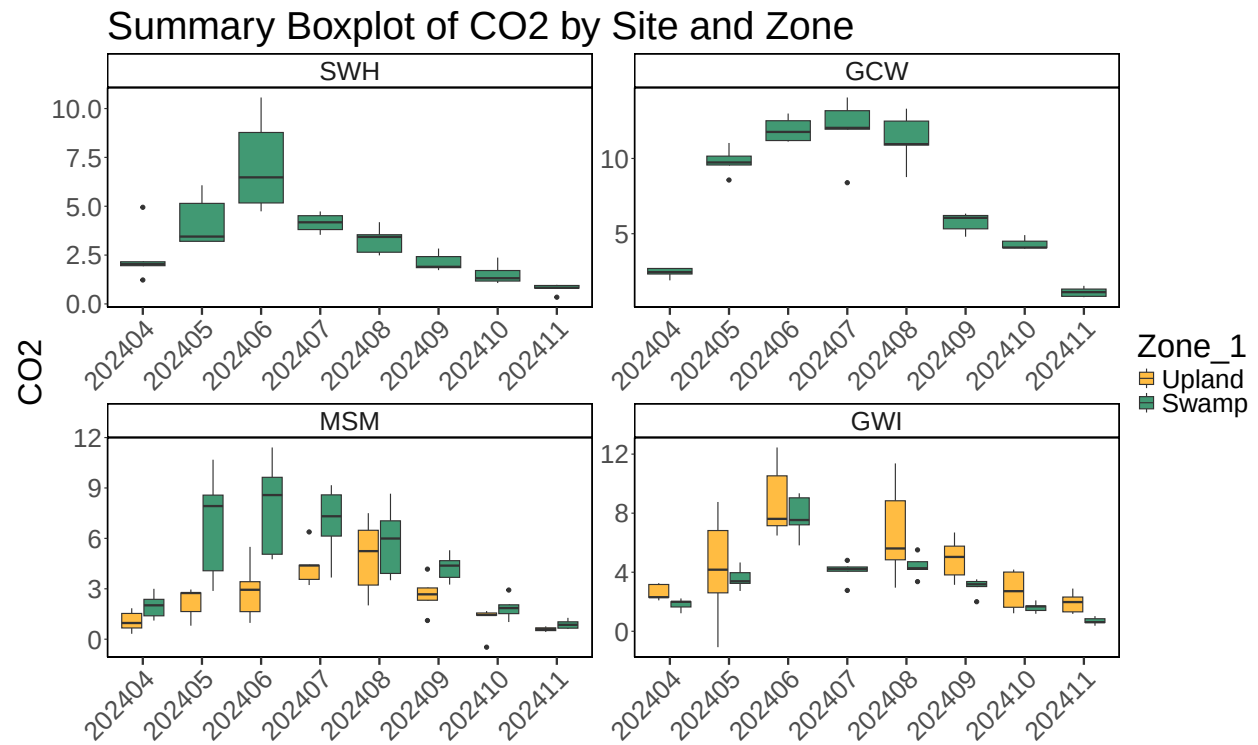
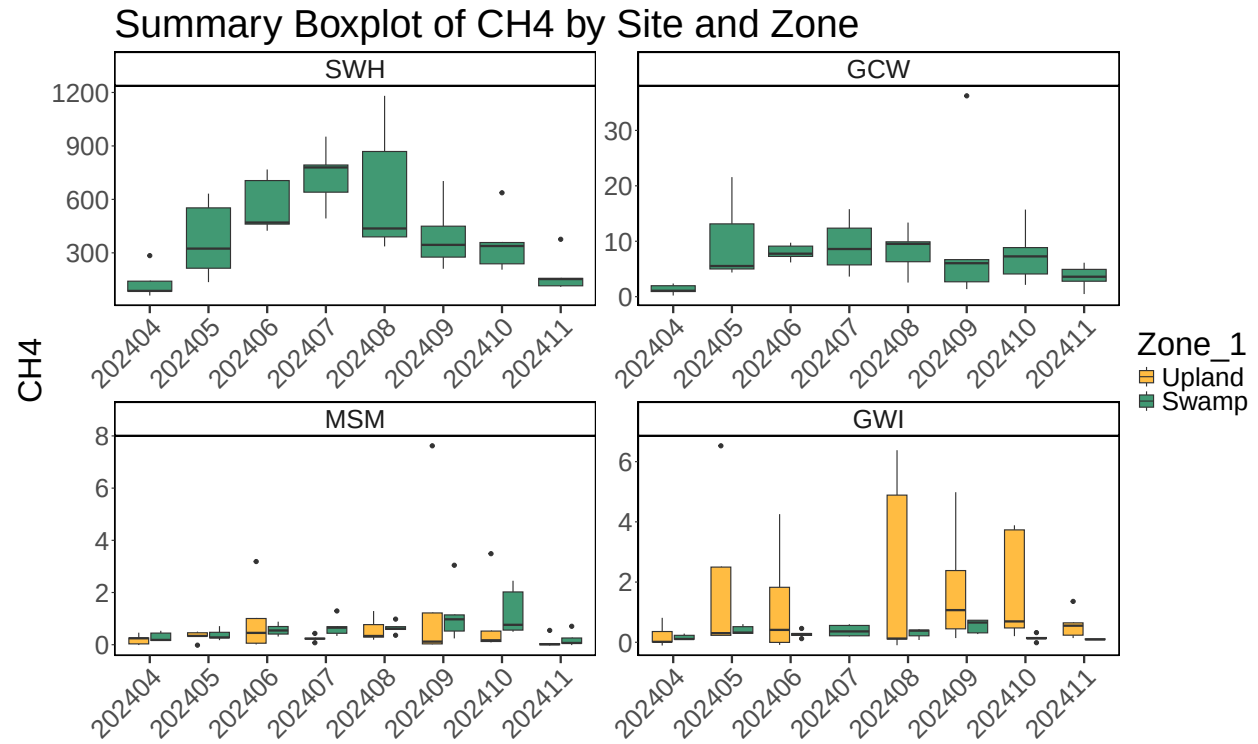
```
all_dat$Zone <- gsub("W", "WC", all_dat$Zone)
```

0.1 Write out files

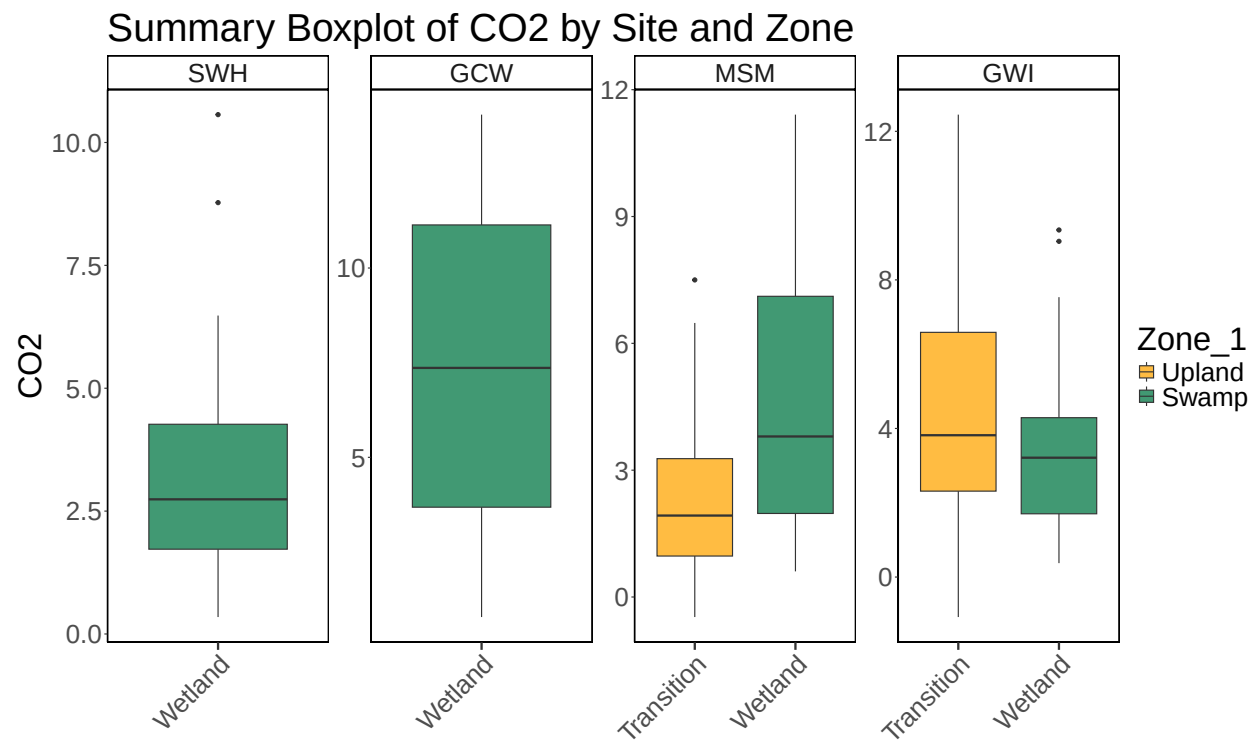
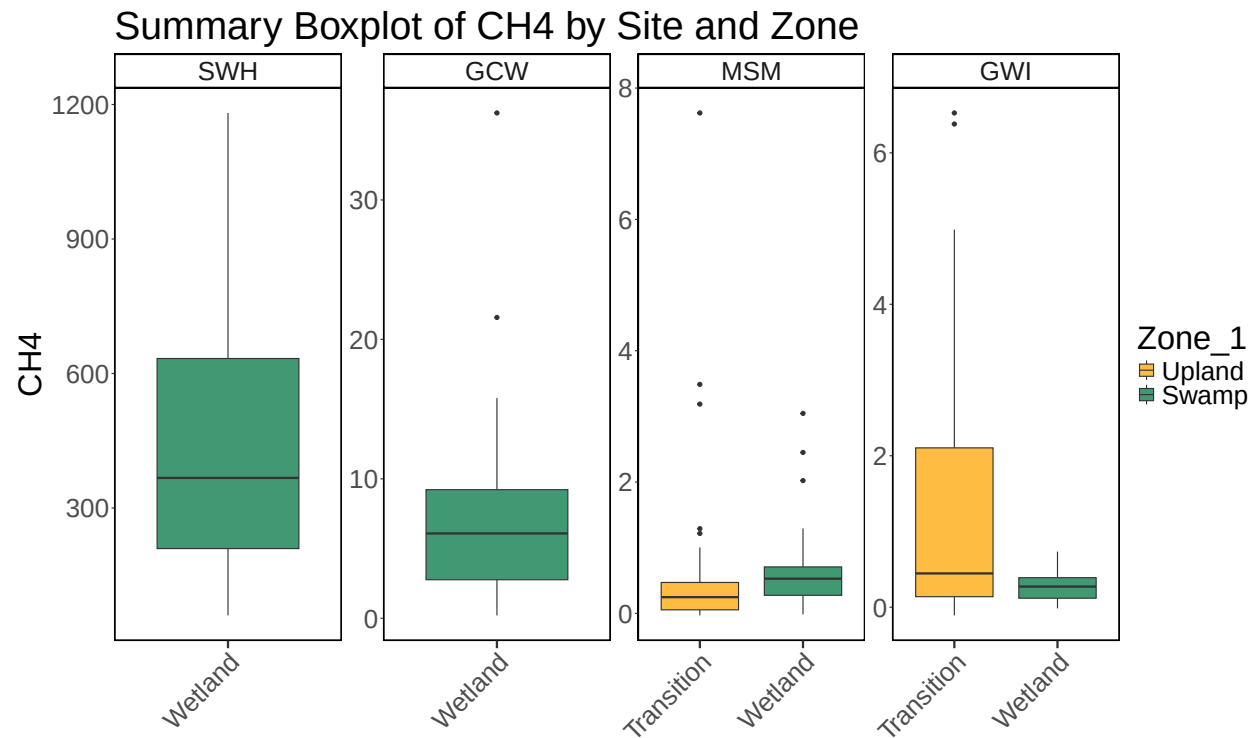
0.2 All Data by Year, Site and Zone



0.3 Boxplot summary by Date



0.4 Boxplot summary by Site



##Calculate summary metrics

0.5 Summary Tables

Table 1: Summary Statistics

Site	Zone_1	min_ch4	max_ch4	mean_ch4	min_co2	max_co2	mean_co2
SWH	Wetland	60.3949819	1181.1091849	422.1361568	0.3444778	10.566791	3.211585
GCW	Wetland	0.2071297	36.2362498	7.3249032	0.7823220	14.050978	7.314527
MSM	Transition	-0.0306176	7.6207208	0.6341064	-0.4740282	7.499684	2.476270
MSM	Wetland	-0.0148283	3.0450899	0.6543798	0.6062537	11.409899	4.562867
GWI	Transition	-0.1078555	6.5260897	1.4174352	-1.0769029	12.449510	4.575282
GWI	Wetland	-0.0146008	0.7340419	0.2887976	0.3744852	9.343932	3.372853